Adding (1) and (2) we get

$$AB^2 + AC^2 = BC \times BD + BC \times DC$$
$$= BC(BD + DC) = BC \times BC$$
$$AB^2 + AC^2 = BC^2.$$

Hence the theorem is proved.

DO YOU KNOW?

In India, Pythagoras Theorem is also referred as "Baudhayana Theorem".

Thinking Corner

1. Write down any five Pythagorean triplets?
2. In a right angle triangle the sum of other two angles is _____.

Converse of Pythagoras Theorem

Statement

If the square of the longest side of a triangle is equal to sums of squares of other two sides, then the triangle is a right angle triangle.

Activity 5

(i) Take two consecutive odd numbers.

(ii) Write the reciprocals of the above numbers and add them. You will get a number of the form $\frac{p}{q}$.

(iii) Add 2 to the denominator of $\frac{p}{q}$ to get $q+2$.

(iv) Now consider the numbers $p, q, q+2$. What relation you get between these three numbers?

Try for three pairs of consecutive odd numbers and conclude your answer.

Thinking Corner

Can all the three sides of a right angled triangle be odd numbers? Why?

Example 4.20 An insect 8 m away initially from the foot of a lamp post which is 6 m tall, crawls towards it moving through a distance. If its distance from the top of the lamp post is equal to the distance it has moved, how far is the insect away from the foot of the lamp post?

Solution Distance between the insect and the foot of the lamp post $BD = 8$ m

The height of the lamp post, $AB = 6$ m

After moving a distance of x m, let the insect be at C

Let, $AC = CD = x$. Then $BC = BD - CD = 8 - x$

In ΔABC, $\angle B = 90^\circ$

$$AC^2 = AB^2 + BC^2 \text{ gives } x^2 = 6^2 + (8-x)^2$$
$$x^2 = 36 + 64 - 16x + x^2$$
$$16x = 100 \text{ then } x = 6.25$$

Then, $BC = 8 - x = 8 - 6.25 = 1.75$ m

Therefore the insect is 1.75 m away from the foot of the lamp post.

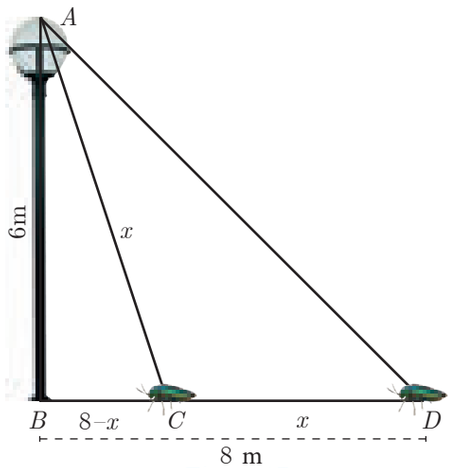


Fig. 4.47

4 Geometry.indd 185 16-12-2020 01:55:11 PM